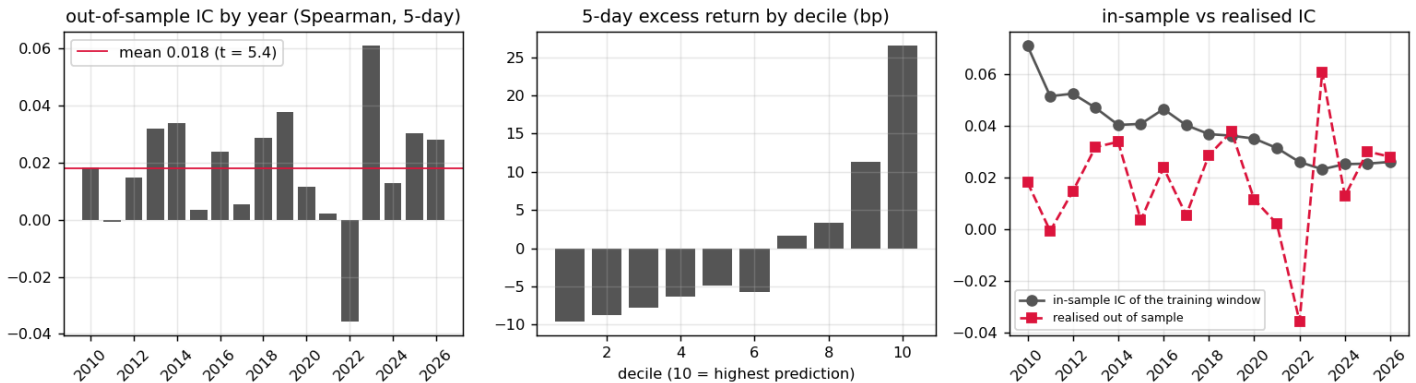


Broker Algo Wheel, Pre-Trade Cost Model and the Research Handoff

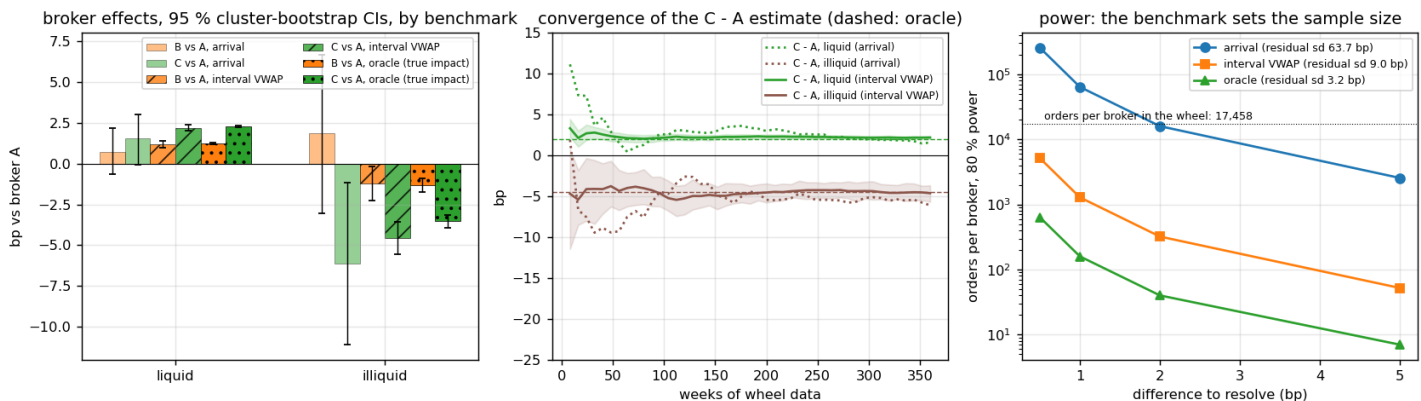
Question. A research team hands a trading desk an equity signal with a gross backtest. What does the desk hand back? Three things, each measured here: which broker algorithm to route to (a wheel run as a randomised experiment, with the sample size the answer needs), what an order of a given size will cost (a pre-trade model fitted on the wheel's fills and tested out of sample), and how much of the signal's gross return survives once the cost model sits inside the portfolio optimiser and the trades go through the brokers.

Method. Free daily data (Yahoo, 156 US large caps, 2006-2026); an ordinary ML signal (eight price and volume features, ridge and gradient boosting, refit each January walk-forward, Grinold alpha); a Ledoit-Wolf risk model; a convex optimiser with dollar neutrality, position and gross limits, a 10 % of ADV desk cap and a transaction-cost penalty from the fitted model. Orders execute in a reduced-form intraday market: a Brownian bridge from the real open to the real close, the real volume on a U-shaped curve, a quoted spread from the liquidity rule, temporary impact concave in participation (exponent 0.6) and linear permanent impact, both at Almgren et al. (2005) scale, and three simulated brokers whose quality differs with liquidity. Because the simulator is ours, every order carries an oracle (its impact-free fill) and every order was also executed with every other broker, so the estimators can be checked against the truth. Phase A (2010-2016) is the desk's calibration: wheel, pre-trade model, cost penalty. Phase B (2017-2026) is the handoff, run once, out of sample.

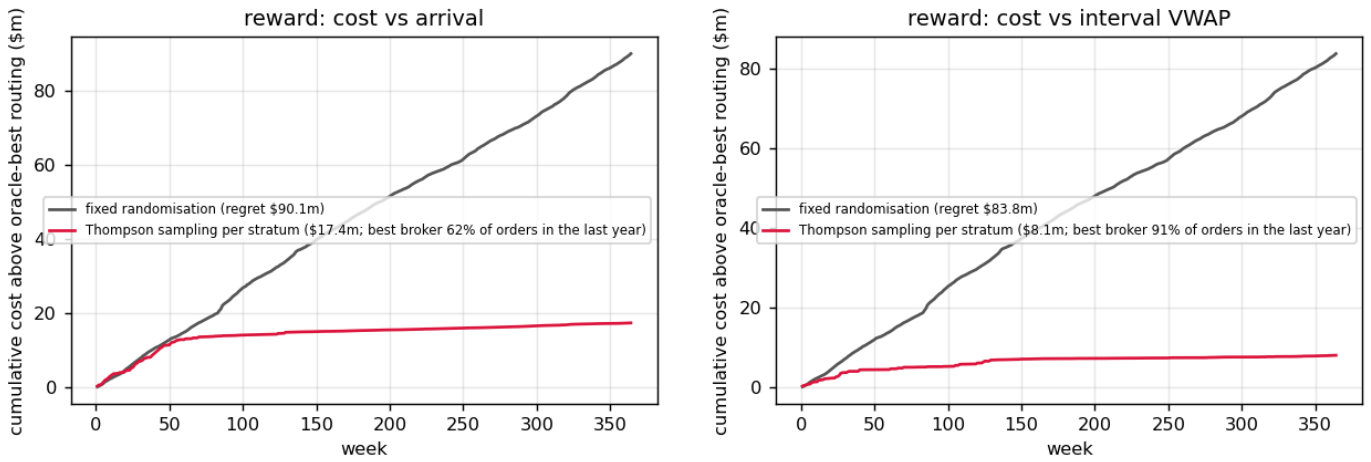
Caveats. Brokers, fills and impact are simulated at published scale; the prices, volumes and the signal are real. The universe is today's large caps, so the signal's gross return carries survivorship bias; the cost results are about the difference between paper and realised books, which the bias affects little. No financing or borrow costs; weekly rebalancing on a five-day signal is deliberately turnover-heavy so that costs matter.



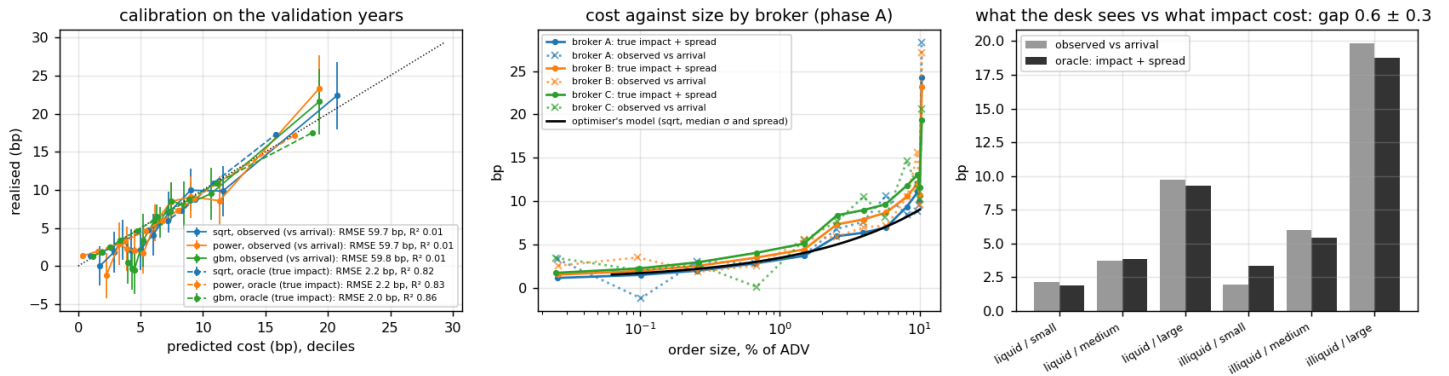
The signal: out-of-sample IC by year, return by prediction decile, in-sample against realised IC.



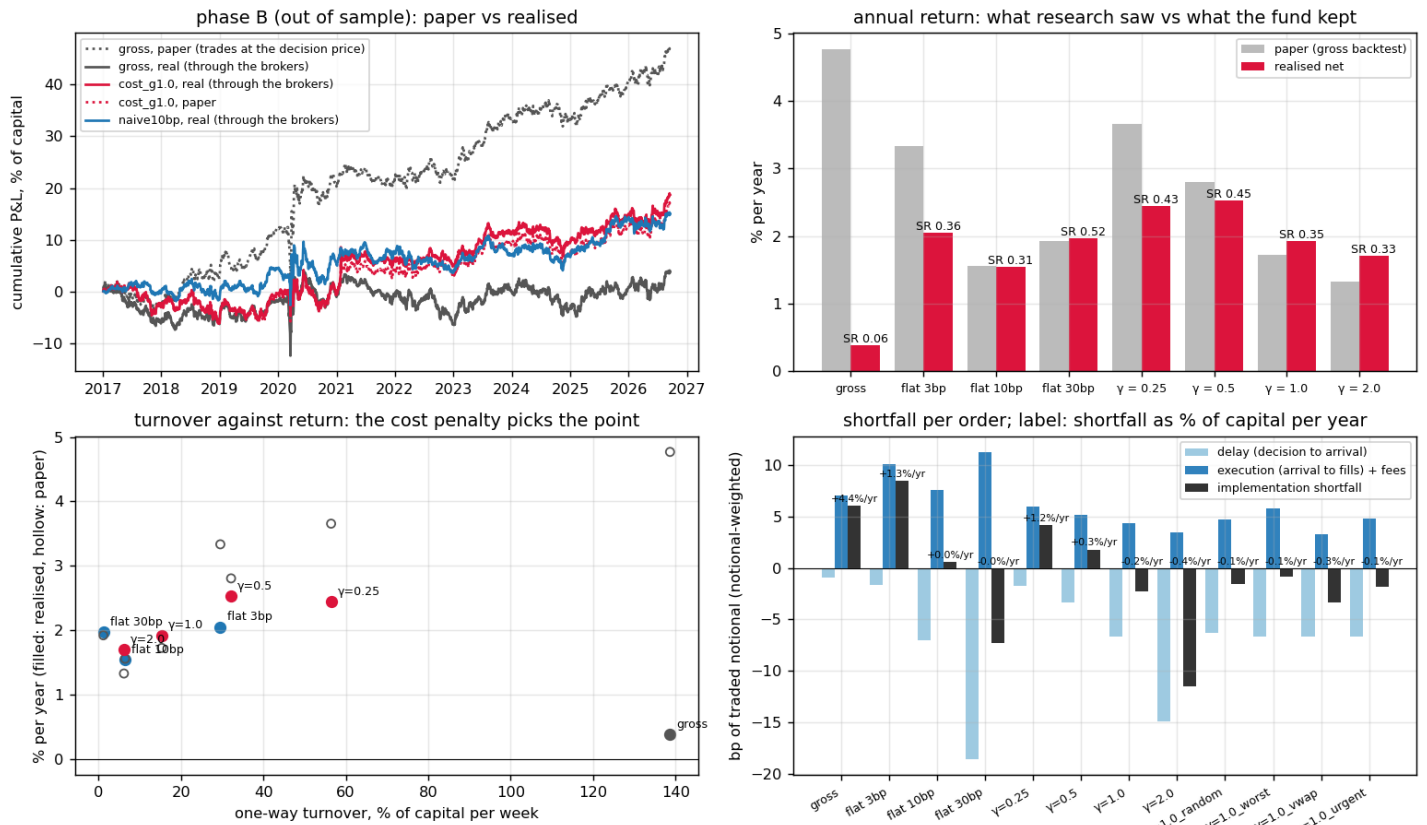
The wheel: broker effects with cluster-bootstrap intervals for three cost benchmarks, convergence of the estimate as weeks accrue, and the orders needed to resolve a given difference.



The cost of learning: cumulative cost above oracle-best routing under fixed randomisation and Thompson sampling, for two reward definitions.



Pre-trade cost model: calibration by predicted decile, cost against order size by broker with the optimiser's fitted curve, and observed against true cost by stratum.



The handoff: paper against realised P&L out of sample, annual return per variant, turnover against return, and the implementation-shortfall decomposition.

ProjectF - broker algo wheel, pre-trade cost model and the research handoff

Generated from results/*.json. Calibration (phase A) 2010-01-01 to 2016-12-31; handoff (phase B) 2017-01-01 to 2026-09-17.

Signal

Out-of-sample Spearman IC of the blended prediction against the 5-day cross-sectional return: mean 0.0178, sd 0.214, t = 5.4 over 4,197 days (ridge 0.0182, gradient boosting 0.0159). Decile 10 minus decile 1: 36 bp per 5 days. Alpha = 0.5 x in-sample IC x sigma x sqrt(5) x z; mean |alpha| 5.3 bp per week.

																	year
IC	0.018	-0.001	0.015	0.032	0.034	0.004	0.024	0.005	0.029	0.038	0.011	0.002	-0.036	0.061	0.013	0.030	0.028

Wheel

Phase A: 52,374 orders over 364 days, brokers dealt by stratified randomisation. True broker parameters: A: eta 0.11 (0.190 illiquid), spread capture 0.75; B: eta 0.142 (0.142 illiquid), spread capture 1.0; C: eta 0.175 (0.085 illiquid), spread capture 1.15.

										target
cost vs arrival (all o	63.7	+0.69 [-0.67, +2.17]	+1.55 [-0.05, +3.02]	+1.85 [-3.06, +6.64]	-6.14 [-11.08, -1.15]	A / C	63,775		15,944	
cost vs interval VWAP	9.0	+1.18 [+1.00, +1.38]	+2.20 [+2.02, +2.38]	-1.20 [-2.25, -0.19]	-4.58 [-5.57, -3.57]	A / C	1,283		321	
oracle: true impact +	3.2	+1.23 [+1.19, +1.28]	+2.29 [+2.22, +2.35]	-1.34 [-1.76, -0.92]	-3.54 [-3.92, -3.16]	A / C	159		40	

Per stratum (interval-VWAP wheel): mean cost by broker and the oracle paired effect.

										stratum
liquid / small	13,366	0.15	1.49	1.93	2.23	+0.46	+0.79	A	A	
liquid / large	25,405	6.79	5.67	7.25	8.58	+1.63	+3.15	A	A	
illiquid / small	456	0.24	2.80	3.32	3.64	+0.50	+0.66	A	C	
illiquid / large	5,103	14.68	19.33	17.34	11.77	-4.22	-9.63	C	C	

Routing map handed over (stratum -> broker; the medium strata are implementation-shortfall orders and inherit their liquidity group): {'0': 'A', '1': 'A', '2': 'A', '3': 'C', '4': 'C', '5': 'C'}; oracle map on the schedule-algo strata: {'0': 'A', '2': 'A', '3': 'A',

'5': 'C'}; the wheel agrees with the oracle in 3 of 4 strata.

Adaptive allocation on the same orders over 7.0 years (cost paid, and regret against oracle-best routing; the wheel-for-a-year column is the fixed wheel's regret after its first 52 weeks, what a desk pays to learn the map once):

								reward
arrival	7.70 bp	6.72 bp	6.34 bp	\$90.1m	\$13.4m	\$17.4m	62%	
vwap	5.89 bp	4.82 bp	4.64 bp	\$83.8m	\$12.5m	\$8.1m	91%	

Pre-trade cost model

target: observed cost vs arrival; fitted on 36,957 orders, tested on 15,417.

							spec
sqrt	a -1.14, b 0.257, beta 0.50	59.79	60.3	0.014	0.12	-0.92	
power	a -0.01, b 0.277, beta 0.61	59.62	60.3	0.014	0.12	-0.96	
gbm	35 trees	59.8	60.3	0.010	0.11	-1.40	

target: oracle (impact + spread); fitted on 36,957 orders, tested on 15,417.

							spec
sqrt	a -0.81, b 0.140, beta 0.50	220.41	6.0	0.823	0.91	-0.25	
power	a -2.95, b 0.115, beta 0.30	220.61	6.0	0.827	0.91	-0.23	
gbm	239 trees	2.0	6.0	0.862	0.94	-0.20	

Alpha contamination: observed cost vs arrival 7.69 bp against true impact + spread 7.11 bp, gap 0.58 +/- 0.28 bp (buys +1.26, sells -0.19); delay cost (previous close to arrival) 0.25 bp.

Square-root fit on observed costs: b = 0.252; on the oracle: b = 0.164.

Optimiser's model: sqrt (validation RMSE sqrt 60.61 vs power 60.65 bp), fitted on 17,458 orders routed to the wheel's best broker: cost_bp = 0.86 + 0.201 sigma_bp (Q/ADV)^0.50 + 0.00 spread_bp.

True out-of-sample test (fitted on all of phase A, applied to the phase B orders of the chosen run):

							target
observed	88.4	88.3	88.4	88.7	-0.000	-0.001	
oracle	1.1	2.3	0.9	7.5	0.323	0.568	

Optimiser's model on phase B orders: RMSE 88.3 bp, bias -0.46 bp (notional-weighted -0.52).

Handoff

gamma chosen on phase A by realised Sharpe: 1.0. Phase A variants (routed to the wheel's best broker):

												variant
gross	7.42	1.58	-0.19	-0.04	4.9	-0.03	150	1.96	52,374	9.7 (1.3 / 8.1 / 0.3)	7.59	3.20

ProjectF - broker algo wheel, pre-trade cost model and the research handoff

naive10bp	4.56	1.05	3.46	0.79	4.4	0.76	15	1.83	11,271	14.4 (5.0 / 9.1 / 0.3)	3.10	0.32
cost_g0.25	6.35	1.37	2.33	0.50	4.7	0.37	83	1.91	49,945	9.3 (1.9 / 7.1 / 0.3)	4.01	1.03
cost_g0.5	5.51	1.20	3.12	0.67	4.7	0.57	51	1.85	47,353	9.0 (2.5 / 6.2 / 0.3)	2.37	0.56
cost_g1.0	4.58	1.02	3.34	0.74	4.5	0.73	25	1.71	42,407	9.4 (3.1 / 6.1 / 0.3)	1.23	0.28
cost_g2.0	2.99	0.73	2.67	0.65	4.1	0.89	10	1.49	34,440	6.1 (3.4 / 2.4 / 0.3)	0.32	0.12

Phase B, out of sample (2017-01-01 to 2026-09-17):

													variant
gross	4.77	0.79	0.38	0.06	6.0	0.08	139	1.80	76,995	6.1 (-0.9 / 6.7 / 0.3)	3.42	1.06	
naive10bp	1.56	0.31	1.54	0.31	4.9	0.99	6	1.43	8,349	0.6 (-7.0 / 7.3 / 0.3)	0.02	0.15	
cost_g0.25	3.66	0.64	2.45	0.43	5.7	0.67	56	1.66	70,183	4.2 (-1.8 / 5.7 / 0.3)	1.24	0.33	
cost_g0.5	2.81	0.50	2.53	0.45	5.6	0.90	32	1.56	63,928	1.8 (-3.3 / 4.8 / 0.3)	0.30	0.18	
cost_g1.0	1.73	0.32	1.92	0.35	5.4	1.11	15	1.38	53,794	-2.3 (-6.6 / 4.1 / 0.3)	0.18	0.09	
cost_g2.0	1.33	0.26	1.71	0.33	5.1	1.28	6	1.10	40,725	-11.5 (-15.0 / 3.2 / 0.3)	0.37	0.04	
cost_g1.0_random	1.73	0.32	1.86	0.34	5.4	1.08	15	1.38	53,794	-1.5 (-6.3 / 4.4 / 0.3)	0.12	0.09	
cost_g1.0_worst	1.73	0.32	1.81	0.33	5.4	1.05	15	1.38	53,794	-0.8 (-6.6 / 5.5 / 0.3)	0.07	0.09	
cost_g1.0_vwap	1.73	0.32	2.01	0.37	5.4	1.17	15	1.38	53,794	-3.4 (-6.6 / 3.0 / 0.3)	0.27	0.09	
cost_g1.0_urgent	1.73	0.32	1.89	0.35	5.4	1.09	15	1.38	53,794	-1.8 (-6.6 / 4.5 / 0.3)	0.14	0.09	
gross_random	4.77	0.79	-0.66	-0.11	6.0	-0.14	139	1.80	76,995	7.6 (-1.0 / 8.2 / 0.3)	5.47	1.06	
naive3bp	3.34	0.58	2.05	0.36	5.7	0.62	30	1.64	26,370	8.5 (-1.7 / 9.8 / 0.3)	1.31	0.34	
naive30bp	1.92	0.51	1.97	0.52	3.8	1.02	1	0.72	2,826	-7.3 (-18.6 / 11.0 / 0.3)	0.04	0.08	